
PHYS 500 - Homework 1-2

Gradient Descent Optimization and Graphics

Analytical mathematics and raylib visualization record

Student: _____

Date: _____

Method summary

For each landscape $f(x,y)$, I compute the gradient $\text{grad } f$ and Hessian H , solve $\text{grad } f = 0$ for critical points, and classify each point using the eigenvalues of H . Numerical critical points for Himmelblau's function were found with Newton iteration from a grid of initial guesses. The displayed coordinates are rounded; the unrounded Newton solutions were checked using the gradient norm.

Gradient descent update: $(x_{t+1}, y_{t+1}) = (x_t, y_t) - \eta \text{grad } f(x_t, y_t)$

Classification rule: positive eigenvalues indicate a local minimum, negative eigenvalues indicate a local maximum, mixed signs indicate a saddle, and a zero eigenvalue makes the second-order test inconclusive.

Visualization key

The raylib screenshots use the following visual layers: the colored translucent surface is $f(x,y)$; contour curves are drawn on the xy plane; blue arrows show the gradient field; colored markers show critical points; and the yellow path shows gradient descent. Green markers are minima, red marks a maximum, magenta marks saddles, white is the current descent point, and yellow is the initial point.

The following pages present the analytical solution for all five landscapes followed by the corresponding labeled raylib screenshot.

Landscape 1 - Rotated anisotropic quadratic

$f(x,y) = 3x^2 + 2xy + 2y^2$

I. Gradient

$\text{grad } f = [6x + 2y, 2x + 4y]^T$

II. Hessian

$H = [[6, 2], [2, 4]]$

III-IV. Critical point and classification

$\text{grad } f = 0$ gives $x = 0, y = 0, z = f(0,0) = 0$.

Point (x, y, z)	Hessian eigenvalues	Classification	$\ \text{grad } f\ ^2$
(0, 0, 0)	$5 - \sqrt{5} = 2.7639; 5 + \sqrt{5} = 7.2361$	local/global minimum	0

Both eigenvalues are positive, so the critical point is a strict local minimum. Because this quadratic form is positive definite, it is also the global minimum.

Figure 1 - Rotated anisotropic quadratic

Raylib visualization screenshot. The in-application panel identifies the landscape and shows the active visual layers, critical points, and gradient-descent state.

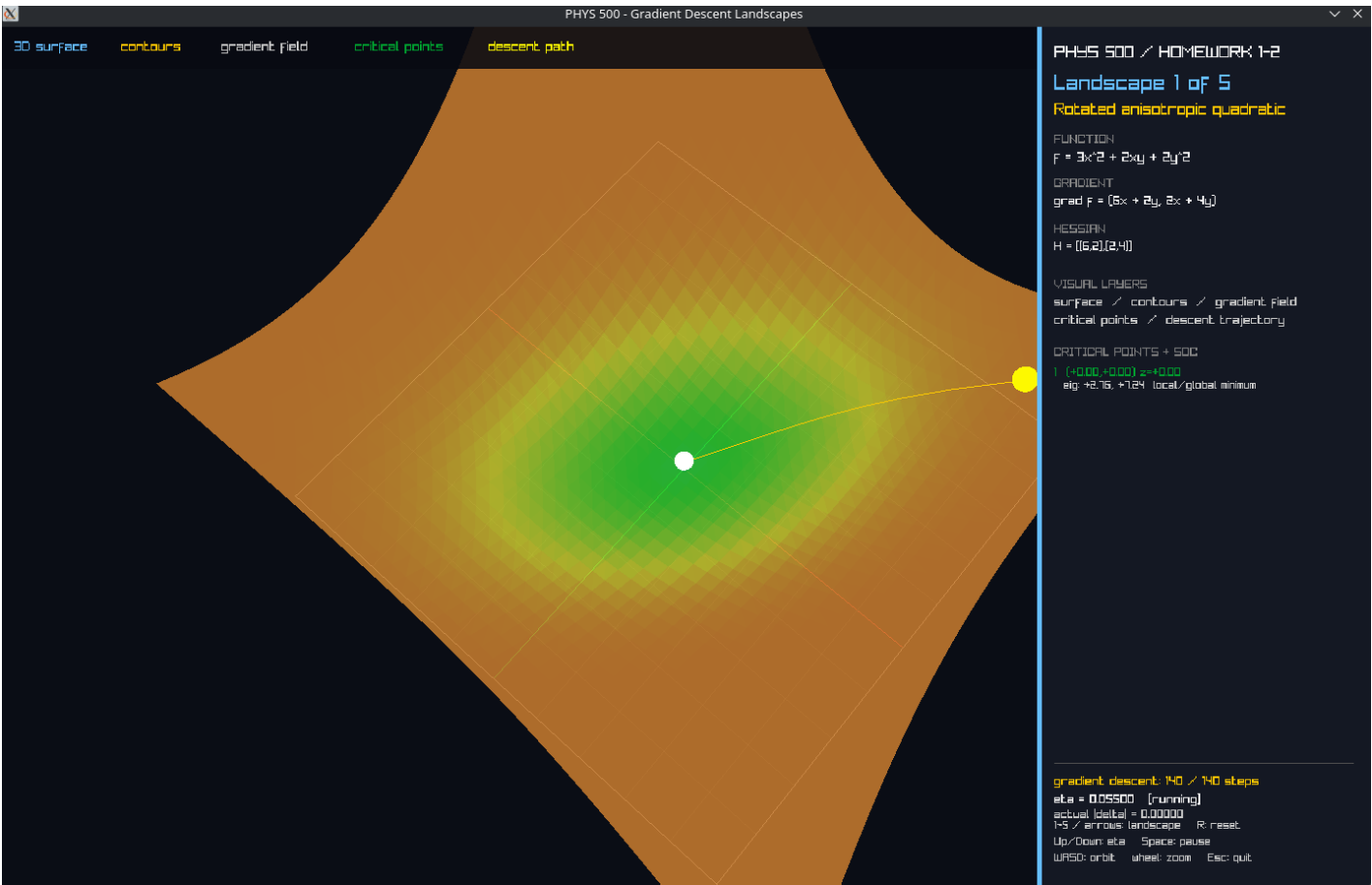


Figure 1. Rotated anisotropic quadratic.

Landscape 2 - Double well

$$f(x,y) = (x^2 - 1)^2 + y^2$$

I. Gradient

$$\text{grad } f = [4x(x^2 - 1), 2y]^T$$

II. Hessian

$$H = [[12x^2 - 4, 0], [0, 2]]$$

III-IV. Critical points and classification

$\text{grad } f = 0$ gives $y = 0$ and x in $\{-1, 0, 1\}$.

Point (x, y, z)	Hessian eigenvalues	Classification	$\ \text{grad } f\ ^2$
(-1, 0, 0)	8, 2	local/global minimum	0
(1, 0, 0)	8, 2	local/global minimum	0
(0, 0, 1)	-4, 2	saddle	0

At $x = \pm 1$ both eigenvalues are positive. At the origin the eigenvalues have opposite signs, so the origin is a saddle point.

Figure 2 - Double well

Raylib visualization screenshot. The in-application panel identifies the landscape and shows the active visual layers, critical points, and gradient-descent state.

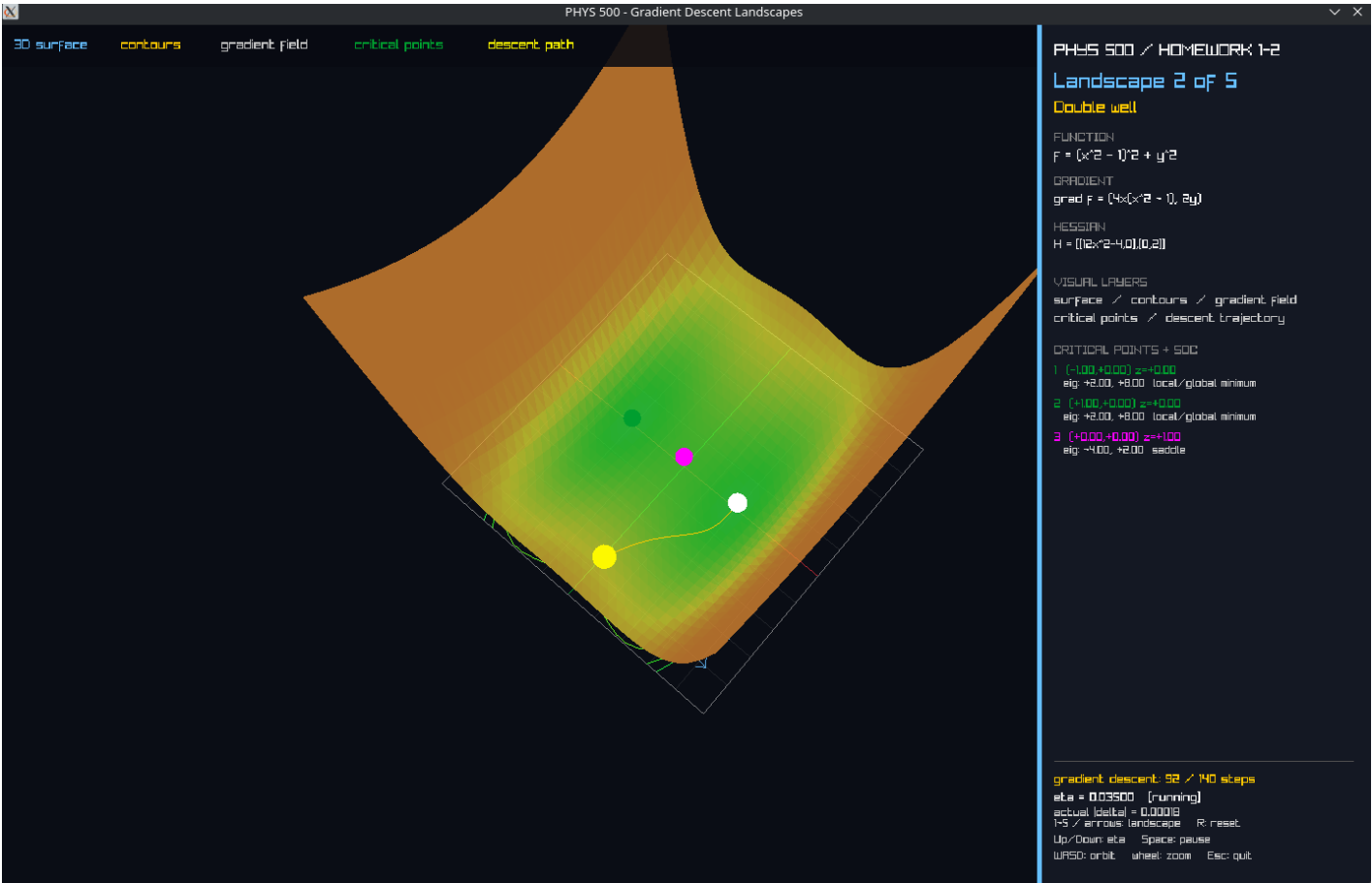


Figure 2. Double well.

Landscape 3 - Lorentzian-Gaussian splatting crater

$f(x,y) = 2 - \exp(-(x^4 + y^2)) - 1/(x^2 + y^4 + 1)$

I. Gradient

Let $A = \exp(-(x^4+y^2))$, $s = x^2+y^4+1$.
 $\text{grad } f = [4x^3 A + 2x s^{-2}, 2y A + 4y^3 s^{-2}]^T$

II. Hessian

$H_{xx} = 12x^2 A - 16x^6 A + 2s^{-2} - 8x^2 s^{-3}$
 $H_{xy} = H_{yx} = -8x^3 y A - 16x y^3 s^{-3}$
 $H_{yy} = 2A - 4y^2 A + 12y^2 s^{-2} - 32y^6 s^{-3}$

III-IV. Critical point and classification

Both gradient components have factors x and y with positive remaining factors, so the only critical point is $(0,0,0)$.

Point (x, y, z)	Hessian eigenvalues	Classification	$\ \text{grad } f \ ^2$
(0, 0, 0)	2, 2	local/global minimum	0

Since $H(0,0) = [[2,0],[0,2]]$, the Hessian is positive definite and the second-order test gives a strict local minimum.

Figure 3 - Gaussian/Lorentzian crater

Raylib visualization screenshot. The in-application panel identifies the landscape and shows the active visual layers, critical points, and gradient-descent state.

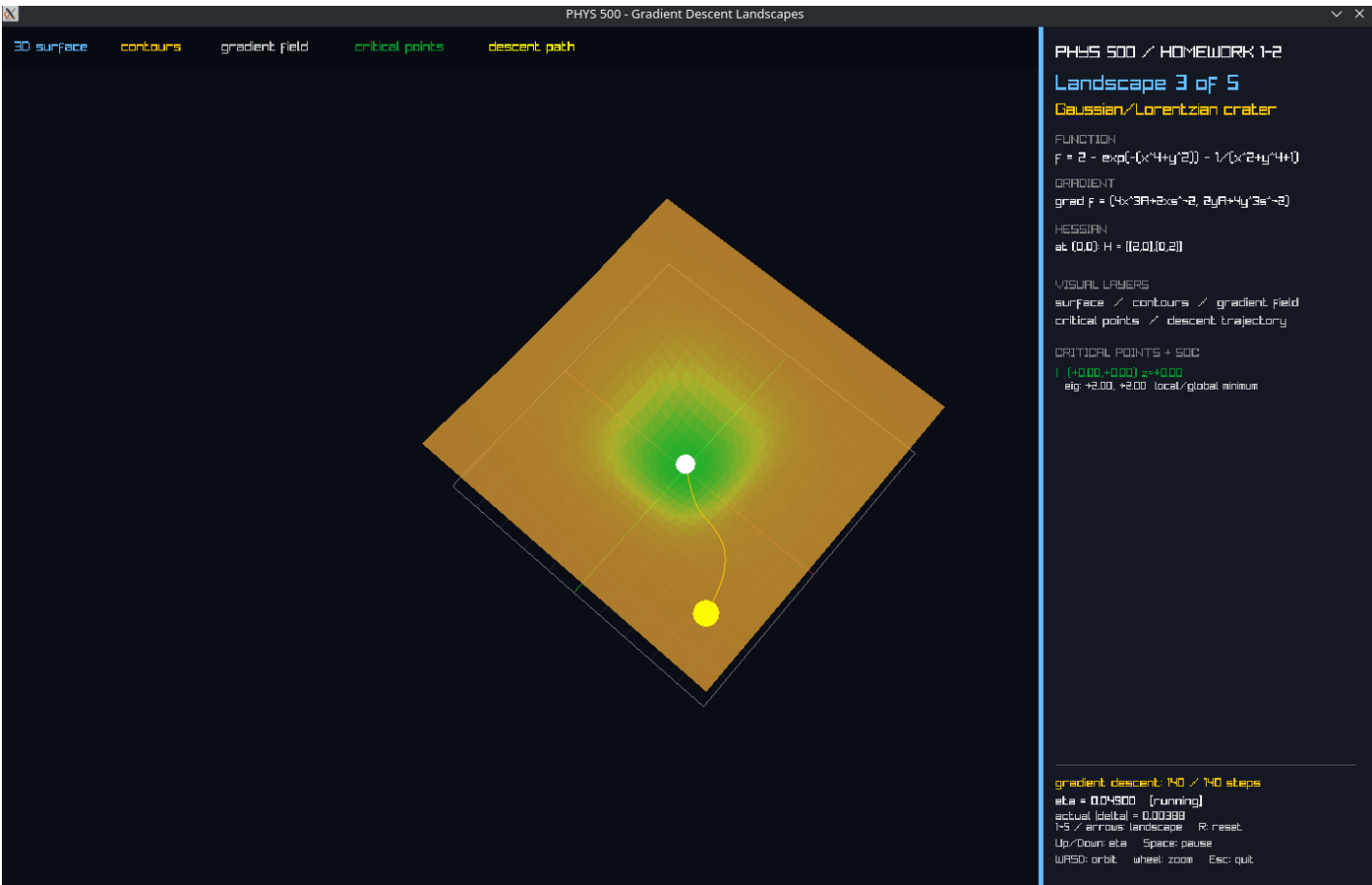


Figure 3. Gaussian/Lorentzian crater.

Landscape 4 - Rosenbrock valley

$$f(x,y) = (1-x)^2 + 100(y-x^2)^2$$

I. Gradient

$$\text{grad } f = [2(x-1) - 400x(y-x^2), 200(y-x^2)]^T$$

II. Hessian

$$H = [[2 - 400y + 1200x^2, -400x], [-400x, 200]]$$

III-IV. Critical point and classification

The second component gives $y = x^2$. Substitution into the first gives $x = 1$, so the point is $(1, 1, 0)$.

Point (x, y, z)	Hessian eigenvalues	Classification	$\ \text{grad } f\ ^2$
(1, 1, 0)	0.3994, 1001.6006	local/global minimum	0

Both eigenvalues are positive, so $(1,1)$ is a strict local minimum. Since f is a sum of squares and reaches zero there, it is also the global minimum. The narrow valley explains why gradient descent needs many iterations.

Figure 4 - Rosenbrock valley

Raylib visualization screenshot. The in-application panel identifies the landscape and shows the active visual layers, critical points, and gradient-descent state.

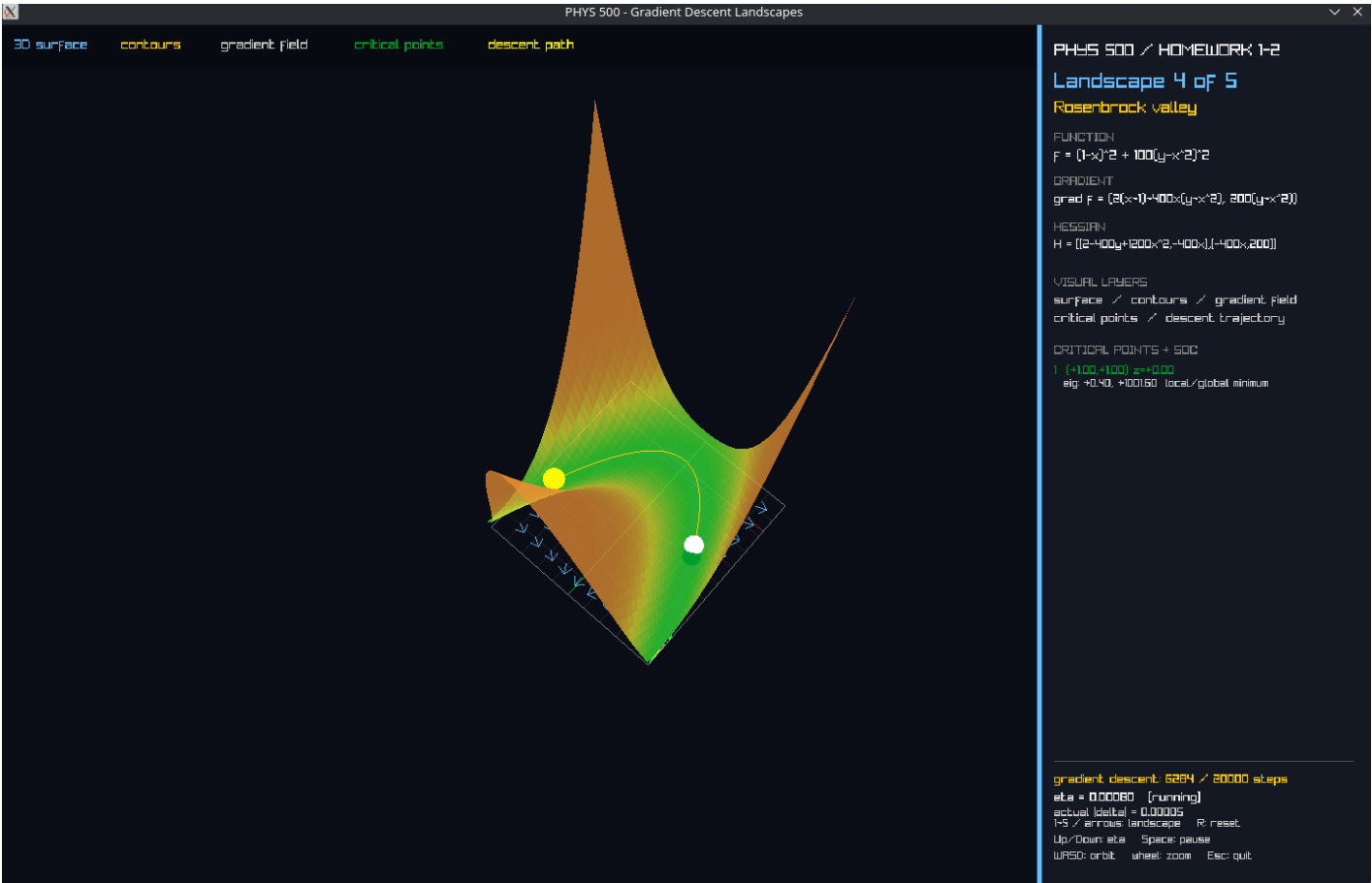


Figure 4. Rosenbrock valley.

Landscape 5 - Himmelblau's function

$$f(x,y) = (x^2 + y - 11)^2 + (x + y^2 - 7)^2$$

I. Gradient

Let $a = x^2+y-11$ and $b = x+y^2-7$.

$$\text{grad } f = [4xa + 2b, 2a + 4yb]^T$$

II. Hessian

$$H = [[12x^2+4y-42, 4x+4y], [4x+4y, 4x+12y^2-26]]$$

III-IV. Numerical critical points and classification

Newton iteration was run from a grid of initial guesses. The table shows rounded coordinates; the reported gradient norms are evaluated at those rounded coordinates. The unrounded solver results were below $1e-8$.

Point (x, y, z)	Hessian eigenvalues	Classification	$\ \text{grad } f\ ^2$
(3.000000, 2.000000, 0)	25.7157, 82.2843	local/global minimum	0
(-2.805118, 3.131313, 0)	64.8404, 80.5501	local/global minimum	3.9e-5
(-3.779310, -3.283186, 0)	70.7144, 133.7856	local/global minimum	3.1e-5
(3.584428, -1.848127, 0)	28.6907, 105.4189	local/global minimum	4.2e-5
(-0.270845, -0.923039, 181.6165)	-45.6052, -16.0660	local maximum	2.3e-5
(-3.073026, -0.081353, 104.0152)	-39.6515, 72.4352	saddle	1.8e-5
(-0.127961, -1.953715, 178.3372)	-50.6102, 20.2841	saddle	1.7e-5
(0.086678, 2.884255, 67.7192)	-31.7066, 75.5076	saddle	3.0e-5
(3.385154, 0.073852, 13.3119)	-14.1352, 97.5479	saddle	1.6e-5

Figure 5 - Himmelblau's function

Raylib visualization screenshot. The in-application panel identifies the landscape and shows the active visual layers, critical points, and gradient-descent state.

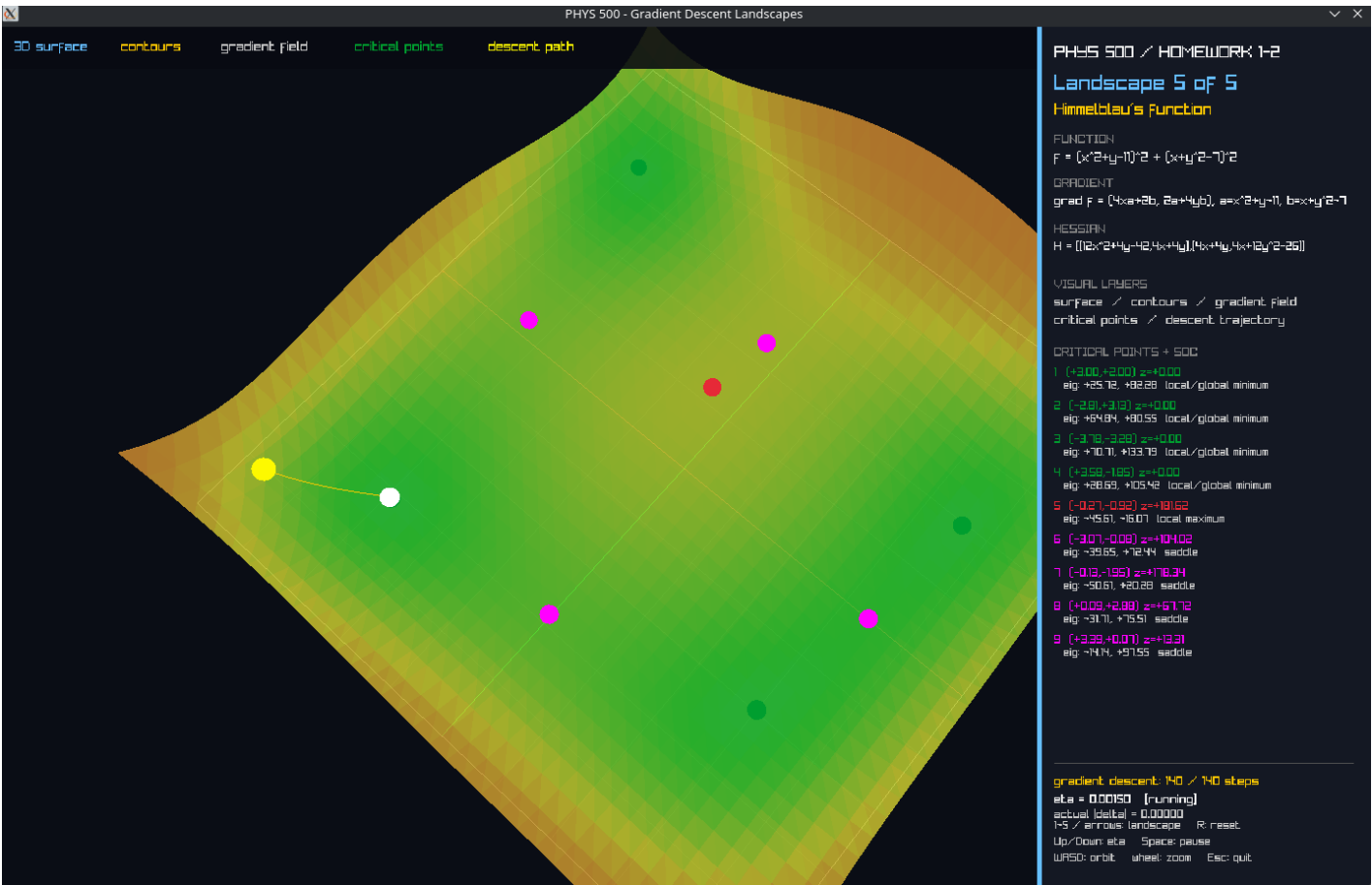


Figure 5. Himmelblau's function.